FNB HACKATHON 2026

UniversityFlow

From Grade 10 to graduation.

A companion for South African learners. It checks every university's own rules, keeps the dates, and stays to the last exam.

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Tumoolo







GRADE 10 UNIVERSITY GRADUATION

CONTENTS

What’s inside

03 The problem, in four numbers

04 One learner’s story: Naledi

05 Why it stands out: ten moments

06 The product, live

07 It thinks ahead: intelligence and safety

08 What learners get next

09 Under the hood: technical architecture

10 The business: who pays, what it costs

11 Try it: demo account, video and code

Try it: universityflow.tumoolo.tech/demo (no password) **Video:** youtu.be/3-Z6S-iAxDI **Code:** github.com/tumoolo-tech/universityflow-fnb-2026

Capable learners, lost for lack of a guide

53.6%

of matric candidates miss a bachelor pass

DBE, 2025 NSC report: 345 857 of 746 110 earned one (46.4%).

25.1%

of Maths candidates reach 50%

477 160 wrote Maths Literacy instead, often chosen in Grade 10, closing doors.

28.6%

finish a three-year degree in three years

DHET, 2015 intake. 20.2% had dropped out by year six.

62.8%

of 15 to 24-year-olds are unemployed

Stats SA, QLFS Q2 2026. 3.8 million are not in work, education or training.

Most are the first in their family to aim for university. Six universities score six ways, every deadline differs, NSFAS is confusing and scams are common. At university, "DP" and "subminimum" are learned the day they cost a year.

Sources: DBE 2025 NSC Examination Report; DHET cohort study (2015 intake, Table 28); Stats SA QLFS Q2 2026. Full list in the long deck and pitch/README.md.

Naledi, 16: the first in her family to aim for university

GRADE 11

Her report, read

She takes a photo of her report. Each mark is read, then confirmed by her. She sees what she is on track for, what is close, and a Plan B.

GRADE 11

A goal she can see

BCom Accounting at UJ. What I need gives the mark to aim for in each subject. Her plan map ticks off each step.

GRADE 12

Applied on time, for free

A checklist and closing date for UJ and Wits, her NSFAS documents ready, and a warning: nobody can sell her a place.

GRADE 12

Through the exams

A countdown to each paper, at most one message a day, and care first on the night she says she is not coping.

JANUARY

The first week

She accepts UJ's offer. A first-week checklist, the campus tour, and her modules checked before the last day to change them.

EVERY YEAR

Every module

How each module is marked, her DP, the exam mark she still needs, free books. Then the next year, to graduation.

Every step above is built and on the live demo. Naledi is made up, and so is everything she enters: try her yourself.

Ten moments that change a learner's path

01

"Your subjects close 35 of 86 courses."

In Grade 10, before it is too late to switch subjects.

BUILT

02

Every university, its own rules

Six ways of scoring, worked out by code: qualify now, close, or Plan B.

BUILT

03

"English: 53% now, 60% needed."

What I need: the exact mark to aim for, subject by subject.

BUILT

04

There is always a Plan B

All 50 public TVET colleges, and "Is this college registered?"

BUILT

05

Money, without scams

NSFAS and ten bursaries, deadline first. Nobody can sell you a place.

BUILT

06

Care before facts

A crisis check before any AI, in five South African languages.

BUILT

07

The first week, not just the offer

Campus routes, step-free too, and modules checked before the last day.

BUILT

08

"You need 41% in the exam."

DP and subminimum explained before they cost a year.

BUILT

09

From test mistakes to a plan

The study coach finds the topics that cost the most marks.

NEXT: PROPOSED

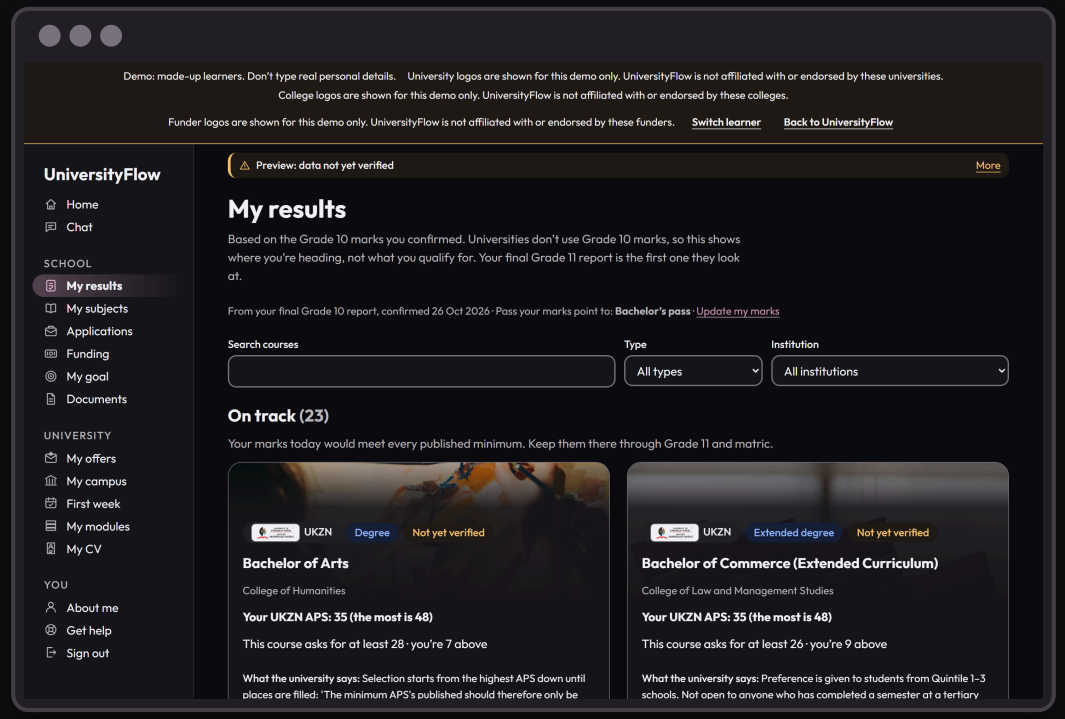
10

From learning to earning

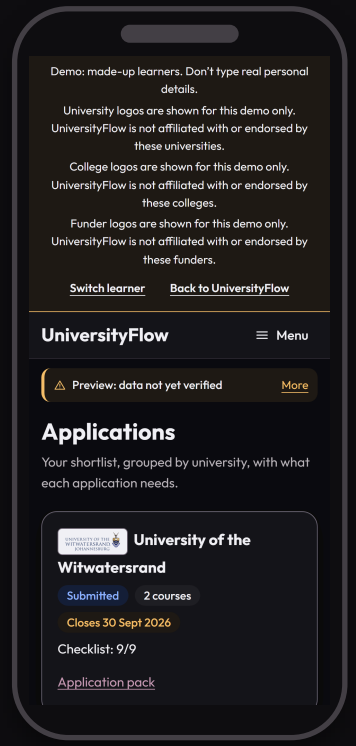
Subject studios, then a CV and portfolio the learner controls.

NEXT: PROPOSED

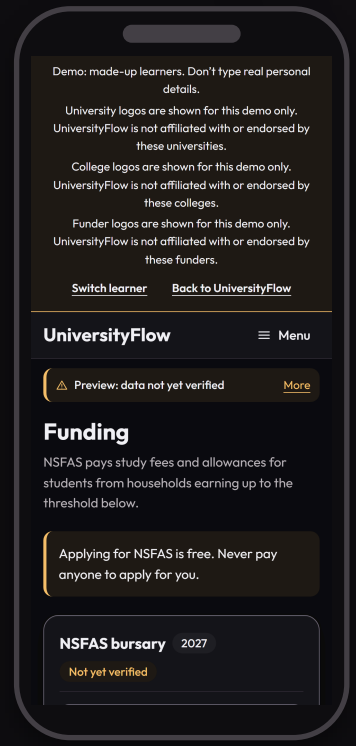
Built and working, on a laptop or a cheap phone



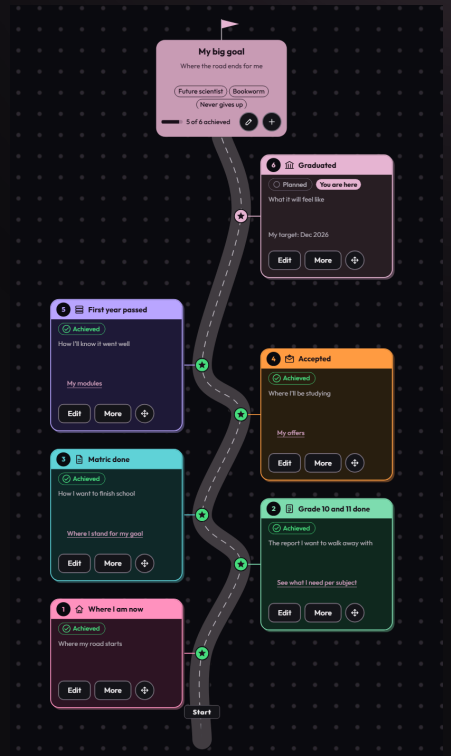
Results, each university's way



Applications



NSFAS, deadline first



Goals they tick off

Also built: What I need · subjects that keep doors open · 50 TVET colleges · ten bursaries · exam reminders · campus and first week · module cards and the mark calculator · Get help

05 · IT THINKS AHEAD

Forecasts from the learner's own marks, explained

AT SCHOOL

Where marks are heading

The pass her marks would give, and each subject's change since Grade 10.

→

Every course, forecast

Qualify now, on track, close or Plan B, each scored the university's own way.

→

What I need

The mark to aim for in each subject to reach her goal.

→

Re-check

After each new report: which courses moved up, which slipped.

AT UNIVERSITY, AND ALWAYS

The exam mark needed

From the year mark so far: DP met or not, the subminimum, what the exam must give.

It remembers

Goals, preferences and notes, kept across years. The learner sees and deletes them.

Knows when to be quiet

Messages fit her exam timetable: at most one a day, never during a paper or after 21:00.

Reads distress first

A crisis check on every message, before the AI, then a person to talk to.

How it thinks: the AI picks among 18 tools, code works out the answer, the AI explains it. Next (Phase 7): a study coach that ranks topics by the marks they cost across tests and their weight in the exam.

Built: packages/admissions (eligibility, school-record, what-i-need, recheck, mark-needed), packages/agent (tools.ts, crisis.ts), packages/messaging (cadence.ts). Phase 7: docs/plans/phase-7-proposal.md.

From the last test, to the next exam, to a first job

The study coach

Phase 7 · planned for Mar 2028

- A photo of a marked test is read, and the learner confirms each mark.
- See the topics that cost the most marks, and how much each carries in the exam.
- A study plan for the weeks before it, around classes, work and family.
- Never a promised mark: an honest starting estimate the learner can change.

NEXT: PROPOSED

The subject studios

Approved plan · planning.md §18.4

- Life Science Studio first, opened straight from What I need for the weak topic.
- One sign-in: the studio gets a pseudonymous id, never a name or phone.
- Progress comes back: “Cells quiz 4/5, 6 of 11 studios explored.”
- Then Maths and Physics, on one shared studio engine.

NEXT: PROPOSED

CV and portfolio

CV built, switched off · portfolio proposed

- A first CV from projects, skills and marks, for jobs and bursaries.
- Small projects tied to each year’s modules, and one degree-long project.
- Progress kept from real evidence, year to year.
- Recruiters see it only if the learner chooses, and only from 18.

NEXT: PROPOSED

Planned, not built: docs/plans/phase-7-proposal.md (the study coach), planning.md §18.4 (the studios, approved 21 Oct 2026), docs/design/portfolio-proposal.md. My CV: packages/cv, off for learners until consent v4.

Technical architecture

Web app

Next.js 16, installable, any browser

Phone app

Expo: Android and iOS, one codebase

SMS and WhatsApp

Reminders, replies, STOP honoured

Data, in South Africa

Postgres and file storage in AWS Cape Town (af-south-1). Row-level security. IDs only in logs. The demo: made-up learners only.

API on Vercel, Cape Town

Sign-in by phone PIN · guardian consent gate · rate limits · security headers. The app never holds a key or calls the AI.

Code decides

admissions: scores, verdicts, What I need, mark maths
catalogue: every record with source, year, checked by
messaging · documents (each mark confirmed) · planner

The AI explains

crisis check first, then the model behind an interface (Claude; Gemini on the demo only), 18 tools, the care charter. Red team: over 450 cases, 100%.

Every provider swappable

AI, SMS, WhatsApp, push, voice, CV (Mmereko), jobs, errors: each behind an interface, with a fake and contract tests.

One TypeScript monorepo · over 5 000 automated tests · guards in CI · 13 audit types per phase · E2E with axe on a production build

Free for learners. Paid for by those who gain.

WHO PAYS

Learners	**R0, always**
Sponsor a learner (CSI, B-BBEE, funders, SETAs)	**R420 a year**
Fee-paying schools, Grades 10 to 12	**R6 000 a year**
Universities, first-year support, from 2028	**R250 000 a year**
Employers, CVs learners share, from 2029	**R2 500 a month**

Free means the app: university fees, tuition and residence stay the learner's, paid by the family or through NSFAS or a bursary.

R10.50 to run, per active learner a month (AI, SMS, WhatsApp, hosting)

2 600 sponsored learners to break even, at about R63 000 fixed a month

R1.8 m Year 1 target: 5 000 learners and 20 schools, against R1.39 m in costs

Why they pay: one NSFAS-funded student costs about R80 600 a year. Keep one learner in 200 from dropping out, and the sponsorship for all 200 has paid for itself.

Estimates, 25 Oct 2026, at R18 to the US$; prices are our proposal, no payer has signed. R80 600: R47.86 billion for 593 543 NSFAS students in 2025 (Minister's reply, NW874).

09 · TRY IT

See it working in three minutes

DEMO ACCOUNT · NO PASSWORD

universityflow.tumoolo.tech/demo

Press **Try as Naledi** (or any of six made-up learners), or **Sign up as a new learner** with any 080 number: the PIN shows on screen.

Demo video (88 s)

youtu.be/3-Z6S-iAxDI

Code

github.com/tumoolo-tech/universityflow-fnb-2026

Full deck (36 slides)

in the repo: pitch/UniversityFlow_FNB_Hackathon_2026.pdf

BUILT THIS WEEKEND

65 changes shipped 23 to 25 October, each through the full test gate: the judge demo, Phase 6 modules and the mark calculator, 50 TVET colleges, ten bursaries, plan maps, My CV.

NEXT

The study coach (marked tests to a study plan) · Life Science Studio in 3D, then Maths and Physics · Mmereko CVs from real projects · pursuing Matric Live and d6 School Communicator.

OUR ASK

People to verify the university data · a legal review of consent for under-18s · an SMS and WhatsApp provider · a first university partner and bursary funders.

Everyone in the demo is made up; university facts are drafts awaiting a person's check. Logos appear as on the demo, with its not-affiliated notice.